Technical report — Gegenprobe: one schema and a converter for Swiss parliamentary answers

Track 2A — OpenParlData · Online · Team 444 — Roman (+ Claude Code agents, §9) · Evidence site https://gegenprobe.mi11ion.io · `make run` re-checks the results (§8) · **Maturity:** census and repair lists = *ready to hand over*; nightly job = *deployable prototype* (one real network night under systemd on a CI runner; not yet running at OPD); rules-only converter = *tested prototype*; review mode, whole and virtual affairs = *feasibility study*; schema v0.1 = *draft*.

1. Summary

The two questions. *Is a common schema feasible?* Yes, with date roles and typed nulls: of 17 fields, **4 are feasible from metadata, 13 partial, 0 need a different concept**; by meaning, 12 are shared, 5 shared with a role split ("answer date" is decided, dispatched or registered), and 3 concepts are body-specific. *Can a converter fill it?* Yes, for what the documents print: shown in depth for the answer date, and for 7 fields of whole affairs, where only the type reaches the unattended bar. Every date and rule value the converter takes from a document carries the page and character span where it is printed; values from OPD metadata are marked `route: metadata` with their source; Apertus's categorical suggestions (type, addressed body, recommendation) are document-level review suggestions without a span; a missing value is a typed null that says why. By default frozen DE/FR/IT rules run without a model; where no rule exists, Apertus points at one of the printed dates as a suggestion a person confirms. **The cross-parliamentary result the schema enables.** OpenParlData (OPD) publishes the affairs of 89 Swiss parliaments with affairs in scope (it scrapes 90). Of 91,268 interventions whose events record a written government answer, **16,749 (18.4%) have no answer text in OPD** (2024–25: 1,855, 14.3%), each typed by cause; 1,008 are Valais oral answers filed as stubs and 406 question-time answers given in the sitting. In Bern the canton publishes 3,532 answers and OPD attaches 229 (6.5%), although 96.1% are already in OPD and linkable; OPD confirmed the bug and that our relink list is the workaround until its fix ships (correspondence, 10 Oct 2026). Deterministic routes propose repairs for 3,107 of the 16,749; 10,800 written answers need per-parliament source adapters. **Where Apertus earns its place** (§6). On 62 documents from 15 parliaments the rules never saw (the 29 answer dates come from 7 of them), Apertus finds 24/29 answer dates where the rules find 2/29; on whole affairs it fills the answer date (22/27 → 27/27; held-out replication 20/35 → 31/35). The local 8B finds the dates too (P2-H2 holds for it) but dates many more answerless documents (§3).

2. Architecture

1. **Census** (`census/`): streams the OPD exports once (metadata only) into a per-affair *answer-link status* (`attached | agenda | parentless | absent | image_only | oral`), the feasibility matrix and a comparability measure; `make run` recomputes it from a vendored extract.
2. **Repair** (`repair/`): deterministic routes per cause, verified by a triple check (affair number printed, answer heading present, decision date = OPD's answer event), as lists keyed on OPD ids; a nightly job (`deploy/nightly/`) re-runs the joins and raises alerts.
3. **Converter** (`convert/`): (i) OPD metadata and events, (ii) frozen DE/FR/IT anchor rules, (iii) the Apertus closed-set pointer on the residue. Modes: `rules_only` (default, unattended, no model) and `review` (suggestions with page, span and margin, flagged `needs_review`); unknown layouts are flagged `no_rules_for_body`.
4. **Schema and validator** (`schema/`, `validate/`): Pydantic schema v0.1, normative; every field is a value with provenance `{value, role, page, char_start, char_end, quote, route, rule, model_build, θ, margin}` or a *typed null* (`not_in_document | unlinked | image_only | oral | text_failed | present_unanchored | no_rules_for_body | not_applicable`); `QAItem` pairs question *n* with answer *n* (not yet filled). The JSON Schema export cannot express the cross-field rules, so validation uses the Pydantic models.

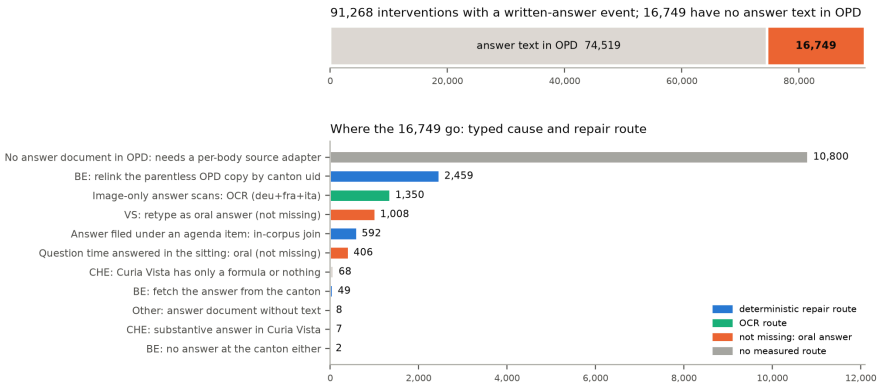


Figure 1. National answer funnel with typed causes and repair routes.

3. Use of Apertus

Closed-set pointer (`convert/pointer.py`, prompts `src/prompts/{de,fr,it}`). The user message holds the document (first 12,000 characters) and the question; the assistant turn is prefilled with a JSON prefix (`{"antwort_datum":`); `null` and every distinct date printed in the document are scored as continuations by summed log-probability (exact forced choice via `prompt_logprobs`). Null if `lp(null) − logsumexp(lp(dates)) > θ`, else the best date, mapped back to its span. θ does not transfer across builds (spike F2: a θ fitted on RLVR scores 0.25 on the final build), so it is fitted per build on dev documents and frozen: θ = 10.40 for the Public AI build (AUROC 0.99, leave-one-out 124/127 [93.3, 99.2]). **Why a pointer** (spikes, `results/report/cited_spikes.json`): asked freely, Apertus dates 15/15 (8B) and 14/15 (70B) Bern interventions that have no answer; a null-allowed prompt fixes the 70B (0/15), not the 8B (14/15).

Which Apertus, which size. Unless marked 8B, Apertus numbers are the **70B** weights, served by Public AI under the name `swiss-ai/apertus-v1.5-8b` (cited: 760/792 greedy outputs byte-identical to the 70B name; MMLU 84.4% vs 70.8% for the v1 8B route); a system message or prefill to the `-70b` name is answered by Qwen-SEA-LION (3/3), so forced choice goes only to the `-8b` name, the response `model` field is checked on every call, and every cached P1/P2 answer comes from one served build (`results/eval/served_models.json`, D10). The **real 8B** (GGUF Q4_K_M, llama.cpp on one 4-vCPU machine, own θ = 16.73; pre-registered exploratory arms, D2/D3) never returns a date that is not printed. On P1 its pointer and cascade score 112/154 and 119/154 vs 130 and 139 on the 70B (McNemar $p = 0.0051$ and 3.6×10^{-5}), and it dates 25 of 90 answerless documents (70B: 7); its residue suggestions are right 7/34 (70B 9/16). On P2 (52 of 62 documents reachable from CI, D2-6) it scores rules 33, pointer 35, cascade 33 of 52 (70B pointer 43, cascade 41; 8B vs 70B 2 vs 10 documents, $p = 0.039$): it finds 15 of 21 answer dates (70B 16, rules 2; P2-H2 holds for it, 13 vs 0, $p = 2.4 \times 10^{-4}$) but dates 11 of 31 answerless documents (70B 4). Its fast scorer failed the parity gate against the frozen scorer twice (57.8, then 4.47 nats), passed a replacement gate written before any 8B arm ran (bf16 reference, 29/30), and its preset self-check failed (10/12); the 8B numbers are fast-scorer numbers (D9).
Recommendation: `rules_only` unattended. For review suggestions, the 8B is the measured sovereign option (one CPU machine; it finds the dates but proposes many more on answerless documents, so a reviewer rejects more); the 70B is the accuracy option on infrastructure the operator chooses (bf16 weights about 140 GB, e.g. two 80 GB GPUs; arithmetic, not measured by us).

Cost (`results/report/cost_end_to_end.json`, one core, no GPU). Per document, fetching politely, extracting and checking takes about 2.1 s, plus 6.9 s per page where OCR is needed; the rules process 480 documents per second (0.5 CPU-hours for all 902,754 documents in OPD's export). Review mode adds the pointer on the residue (97 of 154 held-out documents): 3.5 calls and 5,905 billed prompt tokens per residue document on the 70B, USD 0.59 per 1,000 residue documents at Public AI's list price; the 8B takes 180 s per document on 4 vCPUs (692 CPU-seconds, 9.3 GB RAM), 50 machine-hours per 1,000. On all of OPD's export at that residue share (about 569k documents, an upper bound) the 8B needs about 28,400 machine-hours on one such machine. At ten times OPD's volume: about 5 CPU-hours for the rules, about USD 3,357 for review mode on the 70B.

4. Data and operations

OPD bulk exports (snapshot 6 Oct 2026; OPD's API robots.txt disallows `/v1/`), Bern open data, Fedlex, source PDFs fetched politely (≥ 1.2 s per host; robots.txt per RFC 9309). Both `make run` and the nightly unit run the image read-only (tmpfs `/tmp`), as a non-root uid (10001 when the caller is root), with `--cap-drop ALL` and `no-new-privileges`; PDFs are capped at 50 MB, each pdftotext/tesseract call at 300 s, each document at 600 s, every hit a typed failure row; each event triggers one alert, and failed alerts are re-sent the next night. **One real**

night (`results/monitor/night_real_38062530586/`): under systemd on a GitHub-hosted 4-vCPU runner, one network night against OPD's live exports took 83 min (32 of them downloading 198 MB, about 50 the first night's triple check of all 2,508 Bern rows), 28 min of CPU, 2,078 MB peak, 0 alerts, and reproduced the Bern list on live data (2,459 relink, 49 fetch). Inside OPD (`GP_EXPORTS_DIR`) a night with no changed export files took 96.8 s wall on a copied snapshot with 0 bytes of export transfer; a night re-scanning three days of OPD's changes (80.6% of the docs bytes) is about 21.8 min (derived).

5. Results without a model: schema verdict, census, repair

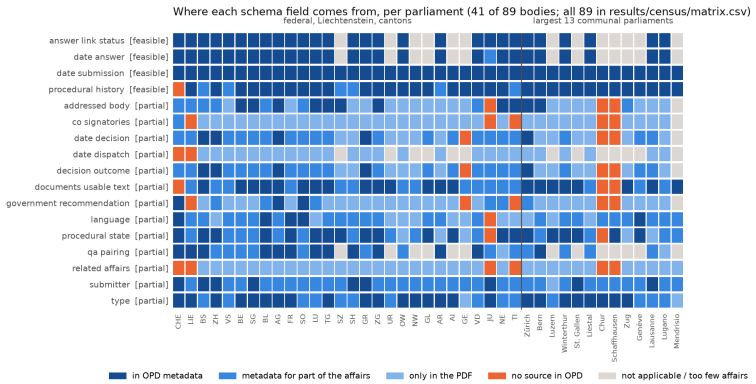


Figure 2. Where each schema field comes from, per parliament (all 89 in `results/census/matrix.csv`). Orange = not in OPD's data for that parliament (the parliament may still publish it), not a different concept.

Where the value lives. A field is *feasible* if OPD metadata carries it in ≥ 80% of eligible parliaments, *fundamentally different* (needing a different concept) if ≥ 20% have no written source, else *partial* (thresholds in `docs/census.md` §3, committed with the first matrix). **4 feasible** (submission date, answer-event date, procedural history, answer-link status), **13 partial**, **0 need a different concept**: no field is orange in a fifth of the parliaments; at a 10% threshold 5 fields would be (`results/census/tier_sensitivity.json`). This threshold measures coverage; the meaning classes below are the finding for the eCH group (`docs/ech-onepager.md`). Type is harmonised for 82.9% of affairs, procedural state for only 278; the government recommendation is in metadata for 34.2% of motions and postulates and PDF-only in 56 bodies.

class	fields and concepts	evidence (`results/census/comparability.json`)
shared	submission date, submitter, co-signatories, addressed body, recommendation, decision and dispatch dates, Q/A pairing, related affairs, language, usable text, link status	one meaning; only where the value lives differs
shared with a role split	**answer date**, procedural history, type, procedural state, decision outcome	in 7 of 54 bodies the "answer" event is mostly the submission date; OPD's event equals the printed date for 37 held-out documents and differs for 7; 35 event labels carry different date roles; 42 of 1,038 type labels map to different harmonised types ("Antrag" alone to 7); 32 state, 3 outcome labels ambiguous
body-specific	oral answers, executive-own business, minutes-only councils	23.6% of question time is oral; 16 bodies publish only meeting documents, 8 without any affair → `virtual` affairs (P4)

29.1% of events have no harmonised id. Where OPD metadata carries a field it mostly agrees with print (P3: type 17/18, submitter 23/28, submission date 15/21; affair numbers 21/29, format differences). Five schema gaps recur across 12 bodies. **Cross-parliamentary question:** across 54 parliaments, 82.3% of written answers have their text in OPD (85.0% after the measured repairs); the median government answers in 68 days (69 without the 7 same-day bodies; Question time 5). OPD's answer date is verified as the decision date in Bern and Ticino (2,379 of 2,379 answers whose title names the date).

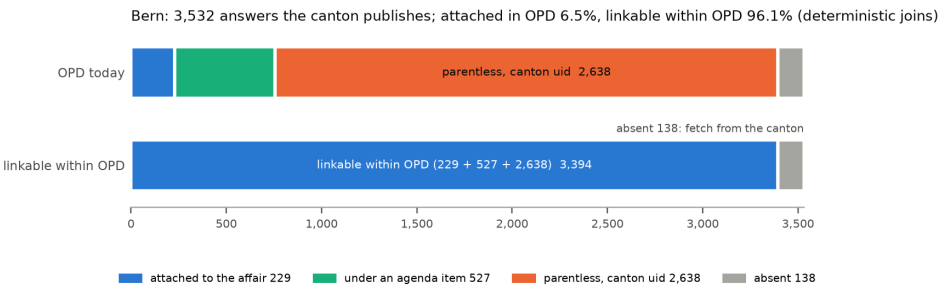


Figure 3. Bern: answers attached to their affair today vs linkable within OPD.

Repair. Bern: 229 attached, 527 under an agenda item, 2,638 parentless with the canton's document uid; 179 parentless answers are not missing (155 affairs already have the text, 24 have no answer event). Checked against OPD's own copy, all 2,459 relink candidates match affair number and heading; the decision date equals OPD's event for 2,332 with the registered parser and 2,421 [97.9, 98.9] with a parser written afterwards (pre-registered sample: 36/40, three parser misses and one canton typo). **Counts per file:** 2,459 relink candidates = 2,421 in `repair_BE_relink.tsv` + 38 in review; 49 fetch candidates = 48 in `repair_BE_fetch.tsv` + 1 in review (`repair_BE_review.tsv`, 39 rows). `python -m gegenprobe.repair.apply_be` writes a dry-run apply plan (before and after state, guarded statements as text). **Note for OPD:** the document `hash` is sha256(body_key + parent_type + url) (49,810 of 49,810 rows of the 10 Oct export), and the 2,421 relink documents are stored as `parent_type = agenda`, so relinking changes their identifier and file URL unless OPD keeps the old one. A second list corrects 727 Bern `begin_date` load timestamps (557 on 2024-10-28 + 170 from later loads, the groups OPD described; 554 from the affair's own submission event). Federal placeholders (57.3% of federal documents): Fedlex recovers 40/50 [67, 88.8]. Scans: in a probe, DE+FR+IT OCR gave usable text for 15/15 [79.6, 100] image-only answers (≥ 200 characters, heading, date: a coarse check an English-only model also passes); where a parliament publishes a text version (Ticino: e.g. `r100.19.doc`), re-taking it is cheaper.

6. Evaluation: five pre-registrations

Design. Every test froze code, prompts, θ and the scorer by sha256 before any label was committed: the LOCK commit precedes the label commit by 6 min 0 s (P1), 2 min 28 s (P2), 3 min 37 s (P3), 4 min 3 s (P4) by commit date (P3b: LOCK5 before the draw), an order GitHub's clock confirms (`results/eval/lock_server_times_all.json`); the annotation files carry no timestamps, so the clock proves order only. Unit: one document's answer date, strict. Arms on identical inputs: **rules**; **pointer** (Apertus on every document); **cascade** (rules, then the pointer on the residue = every review suggestion accepted); **naive** free generation, plain and null-allowed, same weights. **Gold without human labels, by design:** two blind agent annotators (Claude Opus, Sonnet), quotes verified by code, a third breaks ties (order randomised in P2), raw annotations committed, so labels and κ recompute; the committed labels are the record (rerunning the agents need not reproduce them). Non-LLM checks: of 64 P1 answer dates 44 are checkable; OPD's event matches 37 of 44 and 5 of the other 7 are 1–5 days apart (decision vs dispatch); Ticino titles 20/20; P3 answer dates 18/21; the 3 P1 gold dates before OPD's submission date are no gold errors (D11). P2 is thin: 6 of 29 checkable exactly (6/6 agree), 19/29 face at least one non-LLM bound. **Statistics** (post hoc, D10): over the 9 registered confirmatory tests all 4 held conclusions survive Holm (P2-H2 3.8×10^{-6}) and BH; the registered lenient metric changes no registered decision. **Deviations** (`eval/prereg*/DEVIATIONS*.md`): D5/D2-4 naive DD-MM-YYYY re-parse; D6 Branch B as carried out; D7 lock clock; D8 tie-break audit (the Opus tie-breaker sided with the Opus annotator in 36/41 P1 ties, 35 on the recommendation only); D9 8B scorer record; D10 multiplicity, lenient metric, served build; D11 P1 gold audit; D2-5 test population, baselines, θ = 0; D2-6 P2 per parliament on all documents, 8B on P2; D3-5 no P3 dev set; D5-2 a Luzern rule in P3b.

P1 — 154 held-out documents from the parliaments the rules were written for (DE/FR/IT × canton/commune + Ticino scans; 64 with an answer date, 90 without; always-null 90/154; κ 0.95 presence, 0.94 date; 41 tie-breaks).

arm	exact [95% CI]	dates found	wrong printed date	fabricated	values on answerless
rules only (default mode)	**135/154 [81.5, 92]**	47	4	0	2
pointer	130/154 [77.9, 89.3]	46	16	0	6
cascade (suggestions accepted)	139/154 [84.6, 94]	56	11	0	7
8B pointer (CPU, exploratory)	112/154 [65.2, 79.1]	50	37	0	28
8B cascade (CPU, exploratory)	119/154 [70, 83.2]	54	31	0	25
naive, null-allowed	134/154 [80.8, 91.4]	47	11	0	3
↳ null-allowed, re-parsed (D5)	133/154 [80.1, 90.9]	47	13	0	4
naive, plain	98/154 [55.8, 70.8]	47	23	24	39

arm	exact [95% CI]	dates found	wrong printed date	fabricated	values on answerless
↳ plain, re-parsed (D5)	103/154 [59.1, 73.8]	54	26	24	41

8B rows (§3; its registered hypotheses: H1 fails; Branch B, FR+IT+scans rules alone right on 10, cascade alone on 2, p = 0.039; H3 AUROC 0.854). 70B: **H1 null safety failed**: the cascade dates 7 of 90 answerless documents (upper bound 15.2% vs 10%). **H2 gave Branch B** (FR+IT+scans, n = 110: cascade alone right on 5 documents, rules alone on 1, p = 0.22): Apertus adds no null safety (cascade 7/90 answerless dated, rules 2/90), so it left the unattended path (default `rules_only`) and stayed as an opt-in review suggestion (D6), which P2 then tested. **H3 passed**: AUROC 0.969 on all documents, 0.966 on the registered length-matched set. **H4 triggered**: 3 of 8 Apertus-decided FR/IT values are wrong. Post hoc: latest printed date on answer-typed documents 131/154 (vs pointer 11:12 documents, p = 1).

P2 — 62 documents from 15 parliaments absent from every dev and held-out set (29 with an answer date, from 7 of the 15: VD 8, NW 6, Bellinzona 5, OW 4, Vernier 3, SZ 2, Schlieren 1; 33 without; always-null 33/62; κ 1.0).

arm	exact	dates found	wrong printed date	fabricated	values on answerless
rules only	35/62	**2/29**	2	0	0
pointer	**53/62**	**24/29**	8	0	4
cascade	51/62	22/29	10	0	4
naive, null-allowed	52/62	22/29	7	1	3
↳ null-allowed, re-parsed (D2-4)	54/62	24/29	7	1	3
naive, plain	45/62	21/29	8	4	9
↳ plain, re-parsed (D2-4)	49/62	25/29	9	4	9
latest date on answer-typed documents (post hoc)	*44/62*	*18/29*	*13*	*0*	*7*

P2-H1: the rules' recall drops from 47/64 to 2/29. **P2-H2 holds**: on the 29 answered documents, as registered, 22 only the pointer vs 0 only the rules (McNemar $p = 4.8 \times 10^{-7}$; without Bellinzona 17 vs 0, $p = 1.5 \times 10^{-5}$); on all 62, 22 vs 4 ($p = 5.3 \times 10^{-4}$). By parliament (post hoc, D2-1, D2-6): 6 vs 0 with 1 tie on answered documents (sign test p = 0.031), 6 vs 2 with 7 ties on all 62 (p = 0.29). Against the strongest of the trivial baselines we tried, chosen after the results (latest printed date, null on member texts): 10 vs 1 documents (p = 0.012), 6 vs 0 parliaments (p = 0.031). **P2-H3 failed**: the cascade dates 4/33 answerless documents (upper bound 27.3% vs 15%). **P2-H4 holds**: AUROC 0.983. θ = 0, PREREG2's registered secondary: 54/62. *Error profile:* rules right 35, wrong printed date 2, answerless dated 0, missed 25; pointer 53, 4, 4, 1; cascade 51, 6, 4, 1. The cascade trails the pointer because on 2 documents a rule (`rrb_vom`) fired with a wrong printed date, so Apertus was never asked.

P3 — whole affairs: 29 affairs (58 PDFs), 10 parliaments (5 with rules, 5 without), DE/FR/IT. A feasibility study, not a held-out test: extractor, prompts and sample entered in one commit (D3-5). Arms: OPD metadata (M), rules (R), the default metadata + rules (C), C + Apertus suggestions on fields C leaves open (CA); annotators agreed on ≥ 27/29 affairs per field.

arm	type	number	submitter	submitted	answer date	addressed body*	decision no.†
M	17/29	21/29	23/29	15/21	21/27	0/28	0/17
R	26/29	14/29	5/29	4/21	16/27	4/28	5/17
C	**27/29**	**21/29**	**23/29**	**15/21**	**22/27**	**4/28**	**0/17**
CA	27/29	21/29	23/29	15/21	27/27	23/28	0/17

*Every gold value is "executive"; a constant scores 28/28 (D3-3). †Defect D3-1: the frozen cascade's metadata branch drops the rule's number; post hoc C = 5/17. No field reaches the unattended bar (lower bound ≥ 0.80). H3 (Apertus on open fields) 25/32 recall, 25/34 precision is dominated by the near-constant field, so the addressed-body claim is withdrawn; without it 6/8 and 6/9. Set fields (C, registered): co-signatories precision 13/16, recall 13/29 (OPD metadata); related affairs 6/9, 6/14 (rules). **P3b, held-out replication** (PREREG5; 36 fresh affairs, 9 parliaments, D5-1; κ ≥ 0.97 per field): *replicated* by its pre-registered criteria. Apertus on open fields recall 14/16 [64, 96.5]; 4 of 5 eligible fields within or above P3's CIs; the answer date is the field below (C 20/35: in 3 Luzern affairs a frozen rule replaced OPD's correct date, 23/35 had it been kept, D5-2; with Apertus

31/35, its 11 suggestions 11/11 right); precision 14/20 keeps the reviewer gate; type 35/36 reaches the unattended bar.

P4 — virtual affairs: 6 held-out sittings of Bulle's council, 120 agenda items. Rules derive 99 items (precision 99/99, titles 99/99), type 91/99 (H2 holds), the decision for 49/63 (46/63 correct), invent none of 36, and date all 6 sittings (H4 holds). **H1 failed** (recall 99/120, 82.5%; sub-items, unnumbered lines); **H3 failed** (Apertus +2 decisions, p = 0.5). Transfer cell (2 sittings of La Chaux-de-Fonds, D4-4): items 9/9, titles 9/9, decisions 0/9 by rules, 2/9 found with Apertus (1 correct): decision parsing is commune-specific.

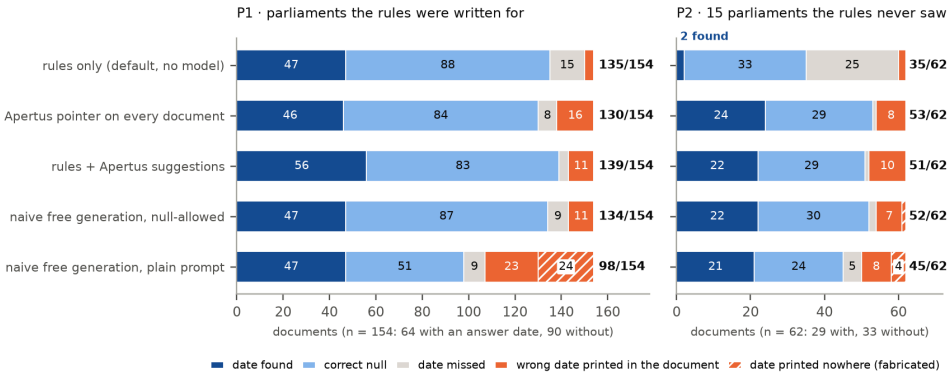


Figure 5. Scoreboards of P1 and P2 (frozen code).

What the tests say. Apertus earns its place where layout varies and no rule exists (P2; the answer date in P3/P3b); where structure is regular (P1, P4) rules suffice. **Why the pointer form:** the same Apertus asked freely with a null-allowed prompt is as accurate (P2 re-parsed 54/62 vs 53/62; discordant 1 vs 0, p = 1), but every pointer value is a printed date with page and span (32/32) and its margin orders the review queue (strong band 26/33 right, weak 3/11, post hoc), while the free prompt quotes its date verbatim for 17/32 values and invents dates printed nowhere (1 in P2; the plain prompt 24 in P1).

7. Limitations

Answer dates are the field tested in depth; P3/P3b are small (29 + 36 affairs), P4 one commune; no gold set contains federal documents; no human labelled a gold value; the 8B was not run on P3/P3b (on P2, Vaud was unreachable from CI). θ is refitted per build; the pointer sees at most 12,000 characters; spans index our own pdftotext/tesseract text (`--opd-text` maps them onto OPD's).

8. Reproducibility

`make run` (Docker, CPU, clean checkout) recomputes the census, the Bern join and triple check, sample conversions with all validator checks, the P1–P4 and P3b scores, the five hypothesis scripts, the 8B analysis and the five LOCK files; re-derives the P1 rules arm from the source PDFs (sha256); recomputes P1 and P2 Apertus decisions from the model cache within a time budget; and re-checks every number in `claims.tsv` against its results file (whole tokens; values of ≤ 3 characters pinned by a context phrase). Locally: rules 154/154, Apertus 154/154 (P1) and 62/62 (P2) identical. In the latest CI run on main whose jobs all passed (38062904003, commit cc37dbd, `results/ci/`): 21/21 replay tests, 142 of 154 rules re-derivations within the budget, all identical; recomputed Apertus decisions matched 129/129 (P1; 25 cache misses from the image's older poppler) and 49/49 (P2; 8 hosts unreachable); 788 claims, 0 failing. That commit precedes the 8B-on-P2 and real-night evidence and this text.

9. Declarations and next steps

Claude Code agents (Anthropic) wrote most code, documentation and the gold labels; numbers are checked by code (`claims.tsv`, lock hashes, verbatim quotes, re-derivation), not by reading. Public AI (which may route outside Switzerland) was used for development and evaluation only. Next: hand the Bern and placeholder lists to OPD and run the nightly job beside their importer; build source adapters for the 10,800 written answers that exist only at the source; date roles, `QAItem` and the five schema gaps in schema v0.2 with the eCH group (`docs/ech-onepager.md`); per-commune decision parsing and sub-item segmentation (the P4 gaps). Report CC-BY-4.0 (`docs/LICENSE-report.txt`). References: OpenParlData exports; Bern Grosser Rat open data; Fedlex; Apertus v1.5 model cards; Popolo, Akoma Ntoso (`docs/mapping.md`).